

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis B: Core antibody, Surface antigen, and Hepatitis D: antibody (HEPBD_L)

Data File: HEPBD_L.xpt

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Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES will provide the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. In addition, NHANES provides the means to better define the epidemiology of other hepatitis viruses. NHANES testing for markers of infection with hepatitis viruses has been used to determine secular trends in infection rates across most age and racial/ethnic groups and has provided a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among people with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in Healthy People and the HHS Viral Hepatitis National Strategic Plan, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Science, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with the NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D and hepatitis E.

An estimated 300 million people worldwide are persistent carriers of hepatitis B virus (HBV). Infection with HBV results in a wide spectrum of acute and chronic liver diseases that may lead to cirrhosis and hepatocellular carcinoma. Co-infection with hepatitis D virus (HDV) in persons

with acute or chronic hepatitis B virus (HBV) infection can lead to fulminant [hepatitis](#). This file addresses HBV and HDV only.

Transmission of HBV occurs by percutaneous exposure to blood products and contaminated instruments, sexual contact and perinatally from HBV-infected mothers to their unborn child.

HBV infection produces an array of unique antigens and antibody responses that, in general, follow distinct serological patterns.

Hepatitis B surface antigen (HBsAg), derived from the viral envelope, is the first antigen to appear following infection and can be detected serologically as an aid in the laboratory diagnosis of acute HBV infection.

Hepatitis B core antibody (anti-HBc) is detectable shortly after the appearance of hepatitis B surface antigen (HBsAg). As the appearance of anti-HBsAg may be delayed after HBsAg clearance, anti-HBc is sometimes the only serological marker for HBV infection and potentially infectious blood. Anti-HBc is found in acute and chronic hepatitis B patients and also indicates past resolved infection.

The Delta antigen/antibody system (HDAg/Anti-HDV) is related to HBV infection but immunologically distinct from its known reactivities; it is the expression of the Delta virus (Hepatitis D Virus, HDV), a cause of severe liver disease in HBsAg carriers. HDV is a 35-37nm particle containing low molecular weight RNA and HDAg, with an outer coat of HBsAg obtained from HBV. HDV is a defective virus and its replication requires helper functions provided by HBV. HDAg has been detected in liver and in serum and induces a specific antibody response (anti-HDV antibodies) in both the IgG and IgM classes.

Tests for anti-HBc, HBsAg, and anti-HDV are conducted as part of the NHANES viral hepatitis component. In August 2025, a new variable (SSLBDHD) was added to reflect retesting of HDV using a different assay after concerns were raised about potential false positive results during original testing (see Description of Laboratory Methodology section for additional information).

Eligible Sample

Examined participants aged 6 years or older were eligible.

Description of Laboratory Methodology

Hepatitis B core antibody (anti-HBc)

The VITROS Anti-HBc assay is performed using the VITROS Anti-HBc Reagent Pack and VITROS Immunodiagnostic Products Anti-HBc Calibrator on the VITROS ECi/ECiQ or VITROS 3600 Immunodiagnostic System.

A competitive immunoassay technique is used. This involves the reaction of anti-HBc in the sample with hepatitis B core antigen (HBcAg) coated wells. Unbound sample is removed by washing. Horseradish peroxidase (HRP)-labeled antibody conjugate (mouse monoclonal anti-HBc) is then allowed to react with the remaining exposed HBcAg on the well surface. Unbound conjugate is removed by washing.

The bound HRP conjugate is measured by a luminescent reaction. A reagent containing luminogenic substrates (a luminol derivative and a peracid salt) and an electron transfer agent is added to the wells. The HRP in the bound conjugate catalyzes the oxidation of the luminol derivative, producing light. The electron transfer agent (a substituted acetanilide) increases the level of light produced and prolongs its emission. The light signals are read by the system. The amount of HRP conjugate bound is indicative of the concentration of anti HBc present.

Hepatitis B surface antigen (HBsAg)

The VITROS HBsAg test is performed using the VITROS HBsAg Reagent Pack and VITROS Immunodiagnostic Products HBsAg Calibrator on the VITROS ECi/ECiQ Immunodiagnostic Systems and the VITROS 3600 Immunodiagnostic System. An immunometric immunoassay technique is used, which involves the simultaneous reaction of HBsAg in the sample with mouse monoclonal anti-HBs antibody coated onto the wells and a horseradish peroxidase (HRP)-labeled mouse monoclonal anti-HBs antibody in the conjugate. Unbound conjugate is removed by washing.

The bound HRP conjugate is measured by a luminescent reaction. A reagent containing luminogenic substrates (a luminol derivative and a peracid salt) and an electron transfer agent is added to the wells. The HRP in the bound conjugate catalyzes the oxidation of the luminol derivative, producing light. The electron transfer agent (a substituted acetanilide) increases the level of light produced and prolongs its emission. The light signals are read by the system. The amount of HRP conjugate bound is indicative of the level of HBsAg present in the sample.

Hepatitis D antibody (anti-HDV)

The International Immunodiagnostics HDV Ab (IID HDV Ab) assay is a competitive enzyme immunoassay (EIA) for the determination of antibodies to Hepatitis D Virus in human serum with a "one-step" methodology. Anti-HDV antibodies, if present in the sample, compete with a virus-specific polyclonal IgG, labeled with Horseradish peroxidase (HRP), for a fixed amount of recombinant HDV protein coated on the microplate. The concentration of the bound enzyme on the solid phase is inversely proportional to the amount of anti-HDV antibodies in the sample and its activity is detected by adding the chromogen/substrate in the second incubation.

From 2007-2008 through 2017-2018, NHANES anti-HDV testing was performed with a DiaSorin enzyme immunoassay designated as Research Use Only in the United States because it was not approved for clinical use by the U.S. Food and Drug Administration. Concerns about high rates of false positivity from this assay were raised after NHANES results showed a higher-than-expected number of positive results for hepatitis D antibody for certain cycles of data collection (Soriano, et. al., 2019). The DiaSorin assay was removed from the market in 2019.

There were no changes to the Hepatitis B core antibody or the Hepatitis B surface antigen lab methods, lab equipment, or lab site for this component in the August 2021- August 2023 cycle.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

Laboratory Method Files

[Hepatitis B Core Antibody](#) (February 2026)

[Hepatitis B Surface Antigen](#) (February 2026)

[Hepatitis D Antibody](#) (February 2026)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials were stored under appropriate frozen (-30°C)

conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line NHANES [Tutorial](#) for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of "0" in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected

during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

The age range and constraints for Hepatitis B and D testing are as follows:

Hepatitis B

The hepatitis B core antibody test is performed on all examined participants aged 6 years and older while the hepatitis B surface antibody test is performed on all examined participants aged 2 years old and older. The Hepatitis B surface antigen is tested only when the Hepatitis B core antibody test is positive. Participant results are coded positive for surface antigen if the surface antigen test is positive; they are coded negative for surface antigen if the test for surface antigen is negative or if the test for hepatitis B core antibody is negative.

Hepatitis D

The Hepatitis Delta Virus (HDV) is a RNA defective virus; and an infection with HDV only occurs in the presence of acute or chronic HBV infection. In NHANES, the test for antibody to HDV is performed on all examined participants 6 years and older who test positive or indeterminate for anti-HBc and HBsAg. The denominator for anti-HDV includes all anti-HBc negative samples, the anti-HBc positive samples that were subsequently found to be HBsAg negative, and the samples that were anti-HBc positive and HBsAg positive that were subsequently found to be anti-HDV negative.

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) is not applicable.

References

- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: <http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx>
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: <https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html>
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: <https://health.gov/our-work/national-health-initiatives/healthy-people>

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR
SAS Label: Phlebotomy 2 Year Weight
English Text: Phlebotomy 2 Year Weight
Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 241728.85724	Range of Values	8068	8068	
.	Missing	0	8068	

LBXHBC - Hepatitis B core antibody

Variable Name: LBXHBC

SAS Label: Hepatitis B core antibody

English Text: Hepatitis B core antibody

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	327	327	
2	Negative	6792	7119	
3	Indeterminate	1	7120	
.	Missing	948	8068	

LBDHBG - Hepatitis B surface antigen

Variable Name: LBDHBG

SAS Label: Hepatitis B surface antigen

English Text: Hepatitis B surface antigen

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	25	25	
2	Negative	7093	7118	
3	Indeterminate	0	7118	
.	Missing	950	8068	

LBDHD - Hepatitis D antibody (anti-HDV)

Variable Name: LBDHD

SAS Label: Hepatitis D antibody (anti-HDV)

English Text: Hepatitis D antibody (anti-HDV)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	0	0	
2	Negative	7118	7118	
3	Indeterminate	0	7118	
.	Missing	950	8068	